AtlasVision X1 multimodal pilot dossier

Research context

Harbor Laboratory operates a small validation line used to test optical sensors under controlled lighting. Before the pilot, technicians recorded observations in free-text notes and attached photographs to separate folders. This process made it difficult to connect a reported defect to the exact image region, device configuration, and inspection step that produced the observation.

AtlasVision X1 was installed above the second conveyor station on 14 May 2026. The device uses three blue optical sensors to capture overlapping views and one amber status light to show that a sample is awaiting review. Lead researcher Lin Yue coordinated the pilot with validation engineer Maya Chen and the laboratory operations team.

Each sample passed through an illuminated inspection zone before reaching the manual review desk. The device produced a product image, a cropped defect region, and a short observation record. The record included the sample identifier, capture time, sensor channel, predicted defect class, confidence score, and a reference to the original image rather than an embedded copy.

The pilot treated the original image as evidence and the extracted statement as a candidate assertion. A candidate could enter the knowledge graph only after its source hash, page or image coordinates, schema type, and review state were validated. This separation allowed the team to correct a conclusion without rewriting the source evidence or losing the earlier decision history.

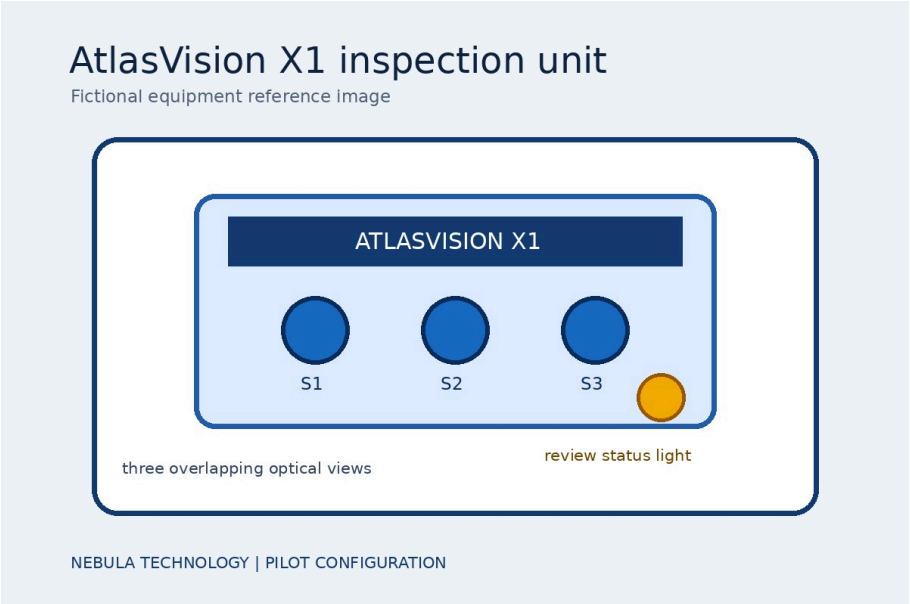
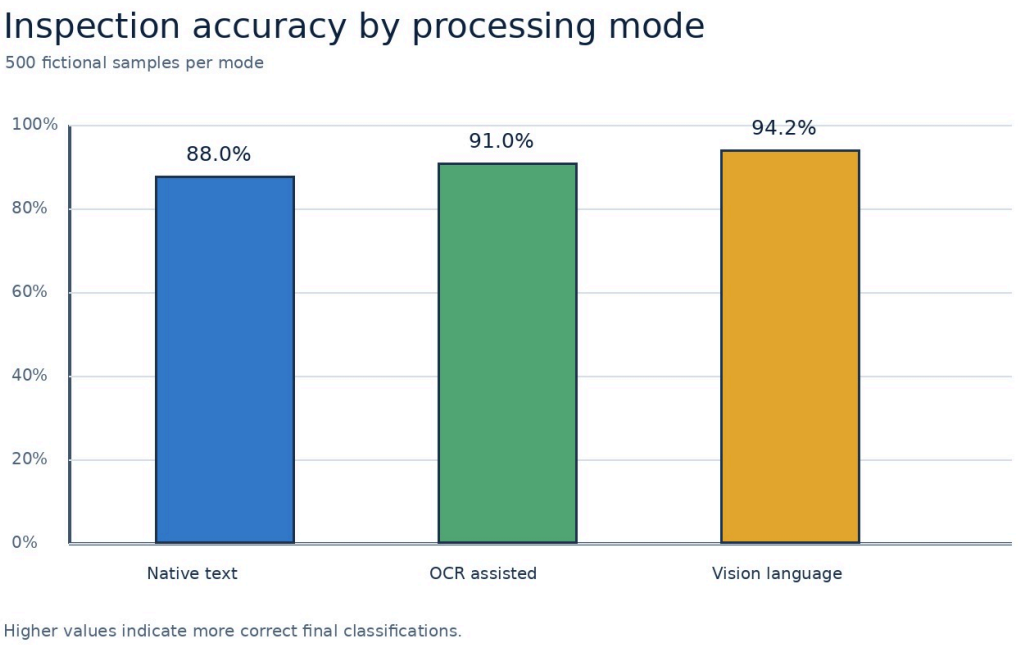


Figure 1. Device configuration and visual regions

The original image remains the evidence object. Extracted entities, properties, and spatial relations are candidates until source hashes, image coordinates, schema constraints, and review state have been verified.

Evaluation results



Mode	Correct	Accuracy
Native text	440	88.0%
OCR assisted	455	91.0%
Vision language	471	94.2%

The native-text baseline classified 440 of 500 samples correctly. OCR-assisted review increased the correct total to 455, mainly because printed batch labels and equipment messages became available to the extraction process. The vision-language workflow classified 471 samples correctly and was strongest on alignment errors that required comparison between the sample edge and a fixture.

Performance gains were not uniform. Reflections on glossy surfaces produced several false positives, and small coating gaps near the image boundary were occasionally missed. The team therefore recommended retaining region previews and review tasks for high-risk observations. The reported 94.2 percent accuracy describes this fictional pilot only and must not be interpreted as a production benchmark.

Deployment and governance

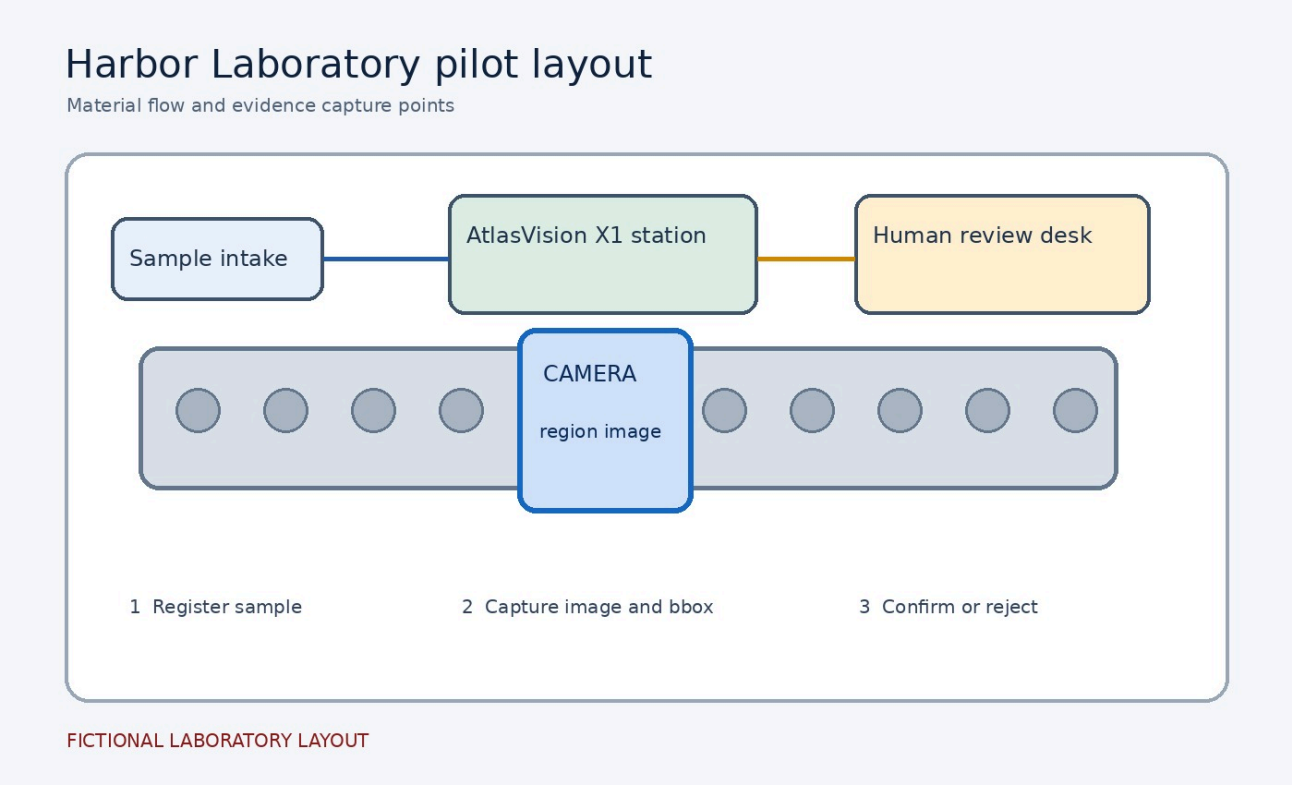


Figure 2. Harbor Laboratory pilot workflow

F1 = 2 * precision * recall / (precision + recall)

The pilot shows how multimodal evidence can improve defect classification while remaining auditable. Nebula Technology will use the findings to refine the prototype and test additional lighting conditions. Any future deployment should preserve the same rule used here: models may propose knowledge, but deterministic validation and accountable human review decide what becomes part of the governed graph.